MindScope: Neurocognitive Modeling, EEG Decoding and Attention Research Platform

Recommended Repository Name:
mindscope-neurocognitive-eeg-research

This name balances discoverability across EEG, neuroscience, cognitive science, computational neuroscience, neural decoding, attention modeling, working memory, machine learning, and brain-computer interface related searches while remaining concise.

Project Vision

MindScope is a research-oriented software platform focused on understanding how attention, working memory, neural representations, and cognitive states can be modeled, analyzed, and decoded from EEG and related neurophysiological data. The platform combines signal processing, machine learning, cognitive modeling, interactive visualization, and research documentation into a single ecosystem.

Core Objectives

- Load and process public EEG datasets.
- Train machine learning models to classify cognitive states.
- Simulate working memory and attention mechanisms.
- Investigate relationships between memory, perception, and attention.
- Explore differences between wakefulness, imagination, and sleep-related states.
- Maintain a research blog documenting findings, literature reviews, and experiments.

Repository Architecture

apps/web – frontend application
apps/api – backend services
packages/eeg-processing – EEG preprocessing pipeline
packages/ml-models – machine learning models
packages/cognitive-models – working memory and attention simulators
packages/shared – shared utilities and types
research/datasets – dataset documentation
research/papers – literature notes
research/experiments – notebooks and experiments
research/results – experiment outputs
blog/posts – research journal and progress updates
docs – specifications, architecture, roadmap

Module 1: EEG Processing Pipeline

Develop a reusable pipeline capable of importing, filtering, cleaning, segmenting, visualizing, and exporting EEG signals. Research Question: How do neural signals differ across cognitive tasks such as motor imagery, attention tasks, and working memory experiments?

Module 2: Neural State Decoder

Train machine learning models to classify cognitive states from EEG data. Initial tasks: • Left vs right motor imagery
• Low vs high working-memory load
• Focused vs distracted states
• Target detected vs target absent Candidate models: Logistic Regression, Random Forest, SVM, CNN, Transformer-based architectures.

Module 3: Working Memory Simulator

Create a computational model of working memory. Features: • Limited capacity
• Memory decay
• Interference effects
• Attention-driven reinforcement
• Memory activation strength Example: A stored representation of a red car increases the probability of noticing similar stimuli in future visual scenes.

Module 4: Attention Modeling

Investigate how active memory representations influence attentional selection. Hypothesis: Objects represented in working memory receive higher attentional priority. Example: Memory contains 'red car'. Visual scene contains multiple objects. The model predicts which object attention is most likely to select first.

Module 5: Dreams vs Thoughts Research

Research-oriented module exploring differences between: • Wakeful cognition • Mental imagery • REM sleep • Non-REM sleep Initial focus: Literature review and EEG-based sleep-state classification. Long-term objective: Investigate whether measurable neural differences can distinguish dreams from internally generated wakeful thought.

Frontend Features

Dashboard
EEG Explorer
Model Training Laboratory
Cognitive Simulation Environment
Research Blog
Experiment Tracking

Backend Features

Data storage for: • Projects • Datasets • Experiments • Model runs • Metrics • Blog posts • Literature notes • Simulation runs

Research Roadmap

Phase 1: Research foundation and literature review
Phase 2: EEG preprocessing and visualization
Phase 3: First neural state classification models
Phase 4: Working memory modeling
Phase 5: Attention simulations
Phase 6: Sleep and dream-state exploration

Long-Term Research Question

Can computational models combining neural signals, working memory mechanisms, and attention systems help explain how internal mental representations influence perception, decision making, and awareness?